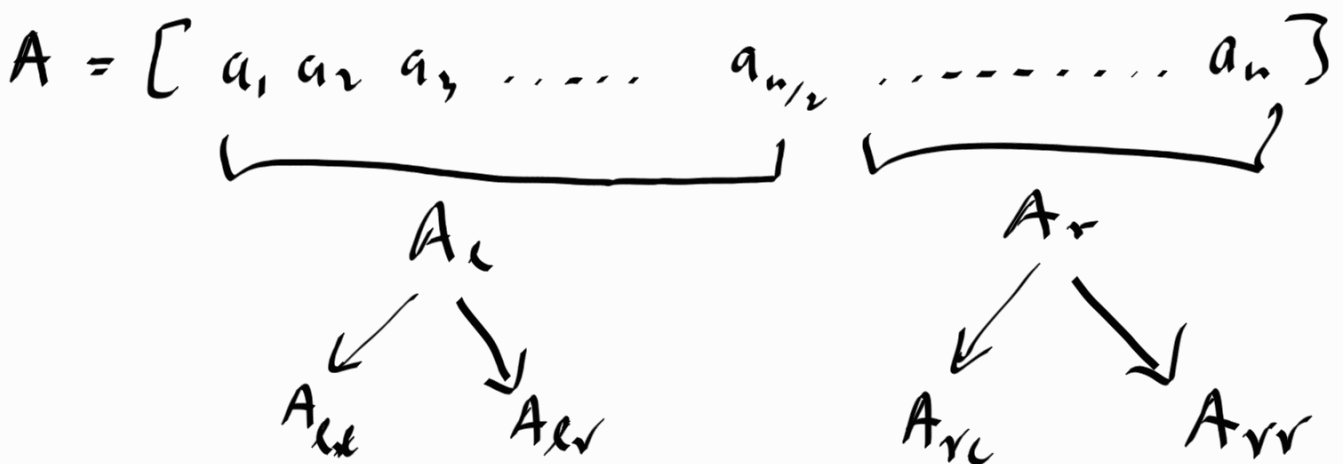


Induction

- Recursion is a common strategy for problems in CS.
 - Take a problem instance
 - Split it into smaller subproblems.
 - Once the smaller subproblems are small enough to tractable, we solve them and combine the solutions.

ex! Binary Search.

- Given a sorted array, we want to find an element.



Mathematical induction is a proof technique that is analogous to recursion.

In general, mathematical induction can be used to prove statements that assert that $P(n)$ is true for all natural numbers.

$$\forall n \in \mathbb{N}: \underbrace{0+1+2+\dots+n}_{\sum_{i=0}^n i} = n \cdot \frac{(n+1)}{2}$$

In induction, we do something like this

Step 1: we prove this claim for the smallest possible case.

(we prove for $n=0$)

Step 2: we assume that this claim holds for $n \geq 0$ and we show that claim is true for $n+1$

$$\forall n \geq 0: P(n) \Rightarrow P(n+1)$$

Let P be a predicate concerning integers greater than 0

$P: \mathbb{N} \rightarrow \{T, F\}$, define $P(n)$ as follows

$$P(n) = \begin{cases} T & , \text{ if } \sum_{i=0}^n i = \frac{n \cdot (n+1)}{2} \\ F & , \text{ otherwise} \end{cases}$$

Let's say that we have proved following 2 things already.

- ✓ 1. The base case ($P(0)$)
- ✓ 2. Inductive case : $\forall n \geq 1: P(n-1) \Rightarrow P(n)$
(step 02)

Let's see why these two steps help us to prove $[\forall n \in \mathbb{N}: P(n)]$

Why?

$P(0)$ ✓

$n=1$

$P(0) \Rightarrow P(1)$ ✓

$P(1)$

P	Q	$P \Rightarrow Q$
0	0	1
0	1	1
1	0	0
1	1	1

we showed that
base case is true.

we proved the
inductive step
and we plugged
 $n=1$

"Modus ponens"

"The way that affirms
by affirming"

since we already
know that $P(0) \Rightarrow P(1)$,
if $P(0)$ is true $P(1)$
must be true.

If $(P(0) \Rightarrow P(1)) \wedge P(0)$,
then $P(1)$ has to
be true

[This is the only case
to consider in this truth
table].

$P(1)$

$P(1) \Rightarrow P(2)$

by using inductive
case and plugging
 $n=2$

$P(2)$

by modus ponens.

$P(2) \Rightarrow P(3)$

by applying $n=3$
to inductive case,

$P(3)$

by modus ponens,

$\vdots$

$\vdots$

$P(n)$

$\vdots$
modus ponens

$P \Rightarrow Q$ is true

P is false

What can you say about Q ?

You can not conclude anything about Q .

Claim: $\forall n \geq 0 : \sum_{i=0}^n i = \frac{n \cdot (n+1)}{2}$

Step 01: Define the predicate $P(n)$

$P: \mathbb{N} \rightarrow \{T, F\}$, $P(n)$ is defined as follows:

$$P(n) = \begin{cases} T, & \text{if } \sum_{i=0}^n i = \frac{n \cdot (n+1)}{2} \\ F, & \text{otherwise} \end{cases}$$

Step 02: State the variable that you are performing induction

- we perform induction over $n \in \mathbb{N}$

we want to show: $[\forall n \in \mathbb{N} : P(n)]$

Step 03: state the base case
 $n=0$ is the base case.

Step 04: prove the base case.

$$P(0) : \underbrace{\sum_{i=0}^0 i}_{\text{LHS}} = \underbrace{0 \times \frac{(0+1)}{2}}_{\text{RHS}}$$

$$\text{LHS} = \sum_{i=0}^0 i = 0$$

$$\text{RHS} = \frac{0 \times 1}{2} = 0$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ base case is true.

step 05: state inductive case:

$$\forall n \geq 1: P(n-1) \Rightarrow P(n)$$

$$\forall n \geq 0: P(n) \Rightarrow P(n+1)$$

$\leftarrow$ This
okay
as well

$$\forall n \geq 1: P(n-1) \Rightarrow P(n)$$

$$\forall n \geq 1: \underbrace{\left[\sum_{i=0}^{n-1} i = \frac{(n-1)n}{2} \right]}_{P(n-1)} \Rightarrow \underbrace{\left[\sum_{i=0}^n i = \frac{n(n+1)}{2} \right]}_{P(n)}$$

Step 06: prove the inductive case.

Assume $P(n-1)$ is true

← Inductive assumption

$$\sum_{i=0}^{n-1} i = \frac{(n-1)(n)}{2}$$

WTS $P(n)$ is true.

consider LHS of $P(n)$

$$\text{LHS of } P(n) = \sum_{i=0}^n i = \left(\sum_{i=0}^{n-1} i \right) + n$$

$$= \frac{(n-1) \cdot n}{2} + n$$

(substituting inductive assumption)

$$= n \left[\frac{n-1+2}{2} \right] = \frac{n(n+1)}{2}$$

↘
RHS of $P(n)$

$$\text{LHS of } P(n) = \text{RHS of } P(n)$$

$\therefore P(n)$ is true.

In other words, whenever $P(n-1)$ is true, $P(n)$ is true.

Step 07: We have shown that $P(0)$ (base case) is true and $\forall n \geq 1, P(n-1) \Rightarrow P(n)$ is true.

$\therefore$, using the principle of mathematical induction

$[\forall n \geq 0: P(n)]$ is true.

Claim: $\forall n \geq 0, \sum_{i=0}^n 2^i = 2^{n+1} - 1$

Sum of first $(n+1)$ terms of powers of 2,

$$\sum_{i=0}^n 2^i = 2^0 + 2^1 + 2^2 + \dots + 2^n$$

$\underbrace{\hspace{10em}}_{n+1 \text{ terms}}$

Step 01: Define the predicate.

$P: \mathbb{N} \rightarrow \{T, F\}$, $P(n)$ is defined as follows:

$$P(n) = \begin{cases} T & , \text{ if } \sum_{i=0}^n 2^i = 2^{n+1} - 1 \\ F & , \text{ otherwise,} \end{cases}$$

Step 02: Define the variable that you perform induction

we perform induction over $n \in \mathbb{N}$

we want to show:

$$[\forall n \in \mathbb{N} : P(n)]$$

Step 03: state the base case,
 $n=0$

Step 04: prove the base case,

$$\text{WTS: } \underbrace{\sum_{i=0}^0 2^i}_{\text{LHS}} = \underbrace{2^{0+1} - 1}_{\text{RHS}}$$

$$\text{LHS} = \sum_{i=0}^0 2^i = 2^0 = 1$$

$$\text{RHS} = 2^{0+1} - 1 = 2 - 1 = 1$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ base case is proven.

Step 05: state the inductive case.

$$\forall n \geq 1: P(n-1) \Rightarrow P(n)$$

$$\left[\forall n \geq 1: \left(\underbrace{\sum_{i=0}^{n-1} 2^i}_{\text{LHS}} = 2^n - 1 \right) \Rightarrow \left(\underbrace{\sum_{i=0}^n 2^i}_{\text{LHS}} = \underbrace{2^{n+1} - 1}_{\text{RHS}} \right) \right]$$

Step 06: prove the inductive case.

Let $n \geq 1$

Assume $P(n-1)$ $\leftarrow$ Inductive hypothesis

WTS: $P(n)$ is true

consider LHS of $P(n)$

$$\text{LHS of } P(n) = \sum_{i=0}^n 2^i$$

$$\text{LHS of } P(n) = \left(\sum_{i=0}^{n-1} 2^i \right) + 2^n$$

$$\text{LHS of } P(n) = 2^n - 1 + 2^n \quad (\text{by substituting inductive hypothesis})$$

$$\text{LHS of } P(n) = 2 \cdot 2^n - 1 = \underbrace{2^{n+1} - 1}_{\text{RHS of } P(n)} \quad (\text{by algebra})$$

$$\text{LHS of } P(n) = \text{RHS of } P(n)$$

Step 07: we have proven $P(n)$ is true when $P(n-1)$ is true $\forall n \geq 1$.

Since we have shown $P(0)$,
and inductive case is true, we
can conclude that $[\forall n \geq 0, P(n)]$
using the principle of mathematical
induction. $\square$